

Vaibhav S

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SUMMARY

Integrated MSc Data Science student specializing in machine learning, reinforcement learning, and AI systems. Experienced building PPO-based RL models, autonomous AI agents, and adaptive RAG pipelines. Focused on scalable, data-driven solutions for analytics and intelligent automation.

TECHNICAL SKILLS

Languages & Databases: Python, SQL, MongoDB, SQLite

ML & AI: Machine Learning, Deep Learning, NLP, Retrieval Augmented Generation, Reinforcement Learning, AI Agents

Frameworks & Libraries: PyTorch, Scikit-learn, Transformers, FAISS, Stable-Baselines3, Gymnasium, NetworkX

Data Engineering & Analysis: ETL, Data Pipelines, Data Warehousing, EDA, NumPy, Pandas

Visualization & Tools: Matplotlib, Seaborn, Power BI, GitHub, PyMuPDF, Groq API, Jira

EXPERIENCE

Data Analyst Intern | AVA Cholayil Health Care Pvt. Ltd.

June 2026 – Present

Tools: Python, Power BI, Excel, Pandas, Matplotlib

- Collaborating with cross-functional teams, including sales, to understand business process and Sales Force Automation (SFA) workflows.
- Analysing real-world business datasets to identify trends and support data-driven decision-making.
- Contributing to dashboards and reports that improve visibility into key business metrics and operational efficiency.
- Applying analytical tools and techniques (Excel, Python, Power BI) to convert raw data into actionable insights

RESEARCH PAPERS

Task-Specific Memory-Augmented Multi-Modal Framework for Fine-Grained Meme Understanding July 2025 – Present

Manuscript in Preparation (2-Member Team) | Tools: PyTorch, FAISS, Transformers, CLIP

- Proposed a memory-augmented multi-modal framework using a FAISS-based memory bank for cross-modal retrieval.
- Implemented memory augmentation to enhance contextual reasoning across vision and text modalities.

PROJECTS

Adaptive RAG for Medical Knowledge Retrieval

Dec 2025 – Mar 2026

Tools: PyTorch, Transformers, FAISS, Python

- Built a medical-domain RAG system with a knowledge-gap detection module to identify missing and conflicting evidence.
- Implemented adaptive query routing across dense, sparse, and hybrid pipelines, reducing hallucinations in healthcare queries.

Autonomous AI Research Gap Detection Agent

Feb 2026

Tools: Python, Groq LLM API, PyMuPDF, SQLite, GitHub Actions

- Built an autonomous agent to monitor arXiv papers daily, extracting tasks, datasets, and metrics using LLMs.
- Designed a gap detection module to flag missing baselines and unexplored datasets; deployed via GitHub Actions with automated email reports.

ParkSmart-RL: Intelligent Parking Slot Selection

July 2025 – Sept 2025

Tools: Python, Stable-Baselines3, Gymnasium, Pygame

- Built a PPO-based RL system on a custom Gymnasium environment (12×3 state space) achieving ~92% accuracy in optimal slot selection after 15,000 timesteps.
- Developed real-time Pygame visualization with vehicle path tracing and reward heatmaps for interpretability.

UFO Sightings Forecasting with Graph Analytics and ML

Jan 2025 – Feb 2025

Tools: Python, NetworkX, Scikit-learn, Pandas

- Analyzed 100K+ global records using graph-based regional similarity modeling; trained a Random Forest model (MAE = 2.04).

- Forecasted 2025–2026 hotspot regions and visualized temporal trends using time-series analysis.

EDUCATION

Amrita University

Coimbatore, Tamil Nadu

Integrated MSc Data Science

Sept 2022 – July 2027

CGPA: 7.31

Relevant Coursework: Machine Learning, Deep Learning, Statistics, Data Visualizations, DBMS, NLP

CERTIFICATIONS

- NPTEL Certified Python for Data Science
- SQL (Basic) Certificate – HackerRank
- Power BI for Data Analysts Certificate – LinkedIn Learning